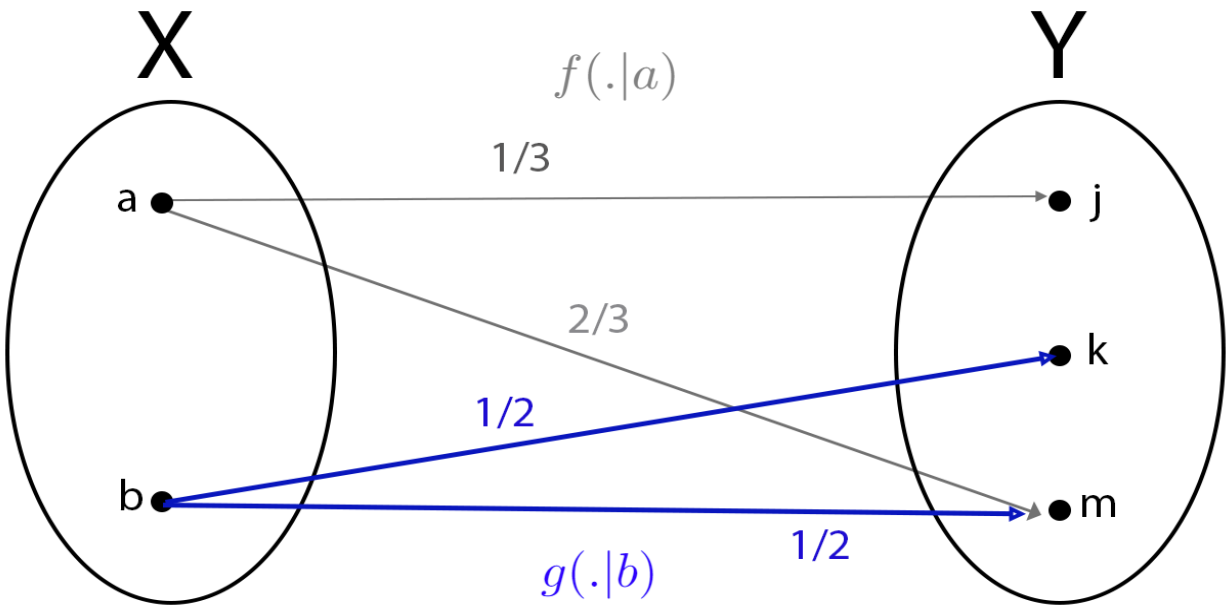
Markov Category : A 10 Minute Introduction

Mahdi Zamani

Nov 2023

Random Morphisms



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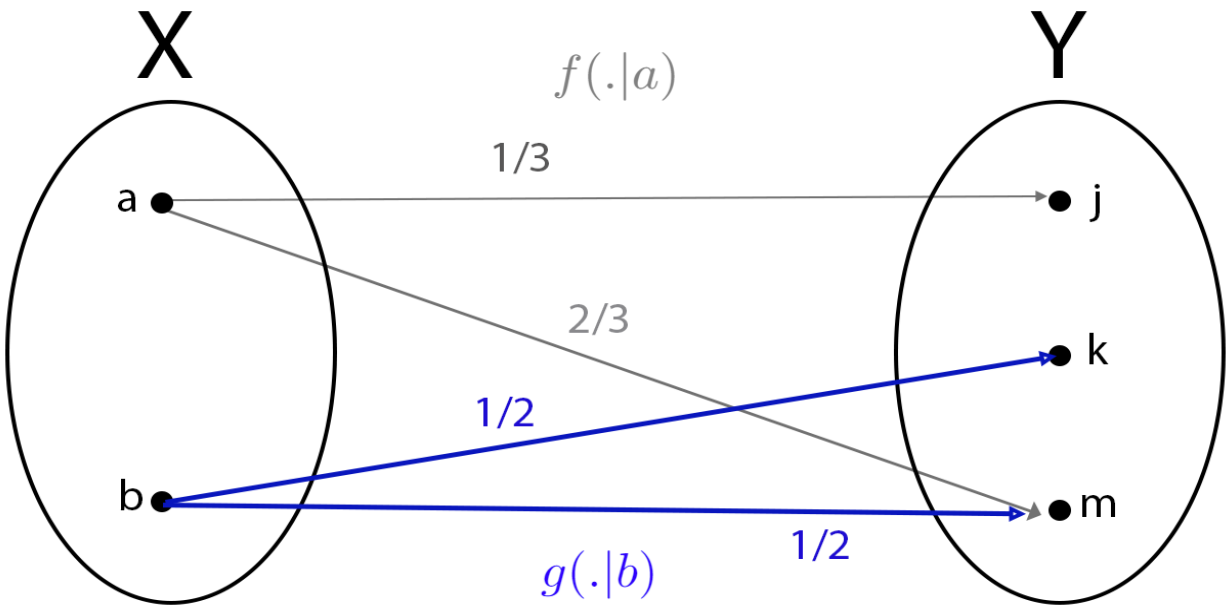
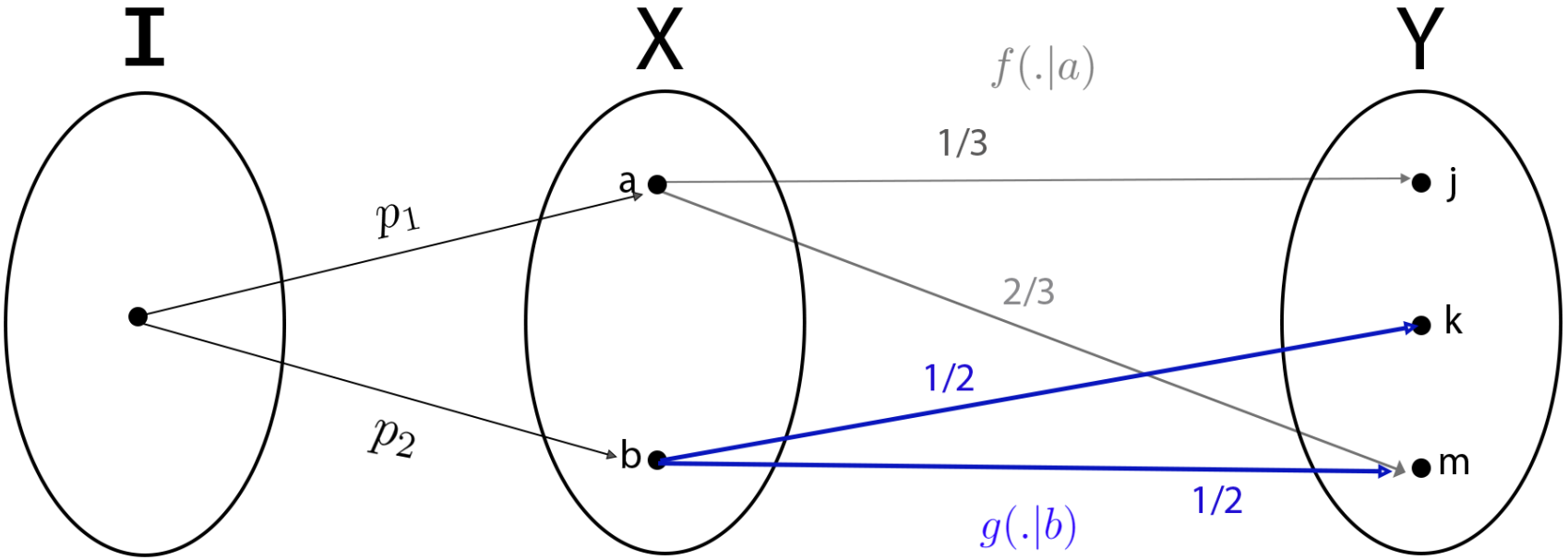


Example

FinStoch Category, with measurable sets as objects, and random morphisms (Markov kernels for Stoch).

$$\begin{array}{cc} \swarrow & \begin{array}{cc} a & b \end{array} \\ \begin{array}{c} j \\ k \\ m \end{array} & \begin{pmatrix} 1/3 & 0 \\ 0 & 1/2 \\ 2/3 & 1/2 \end{pmatrix} \end{array} \qquad f(j|a) = 1/3$$

Probability Distribution



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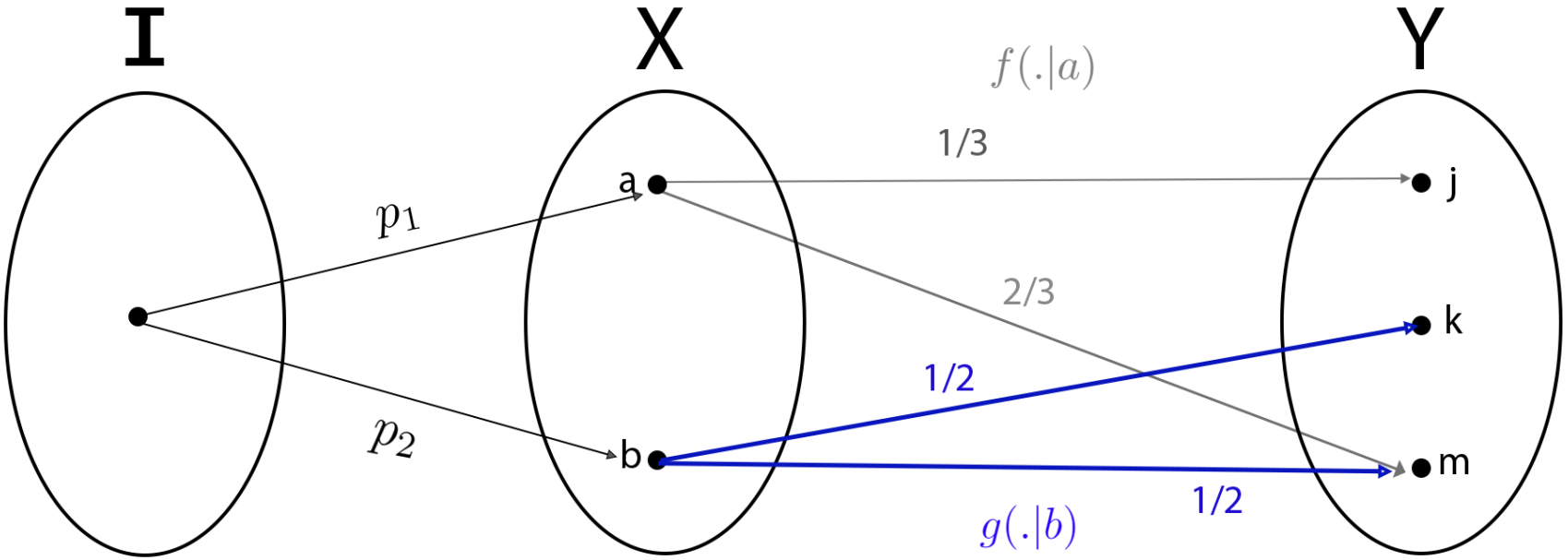
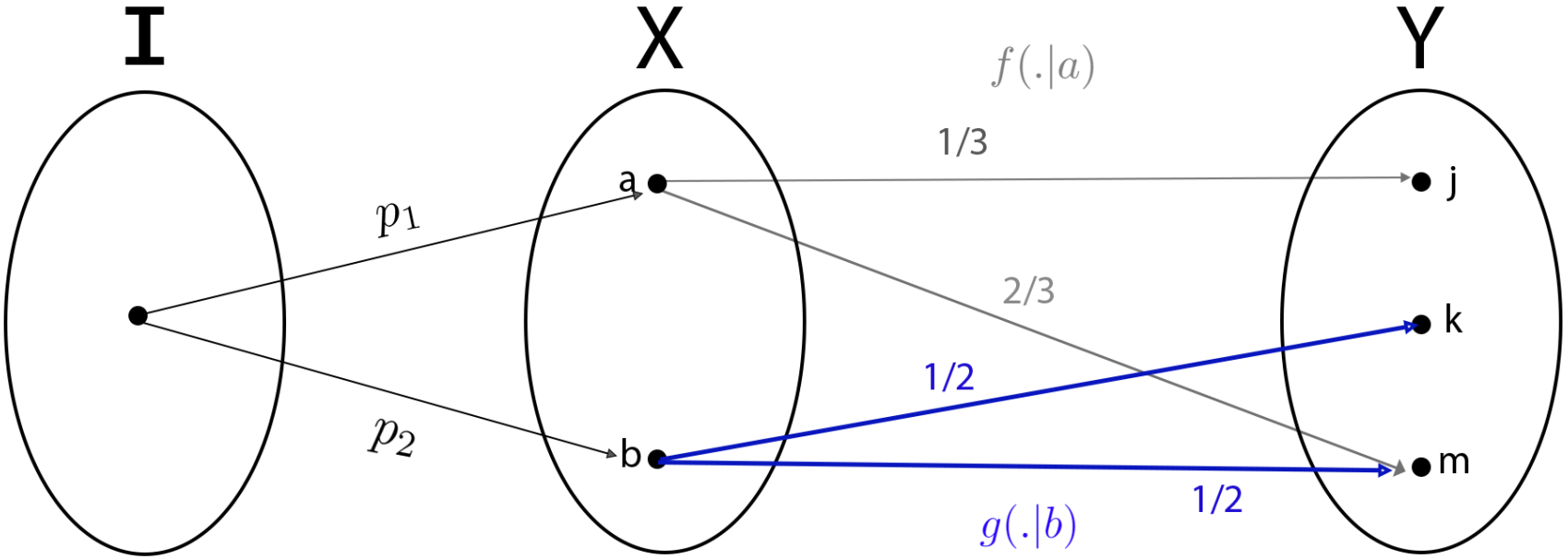


Notation

$$I \xrightarrow{P} X$$

Probability Distribution



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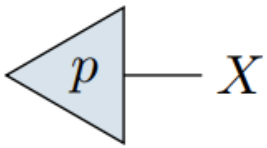

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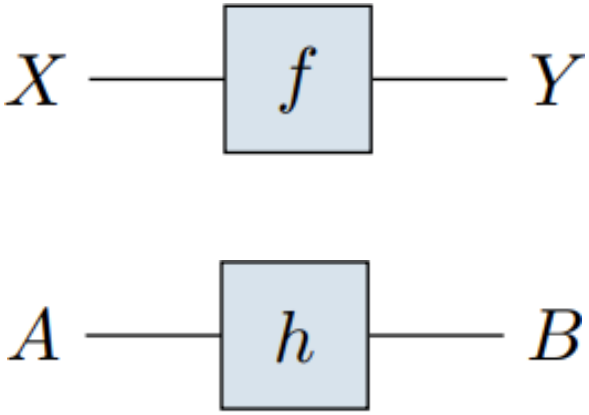

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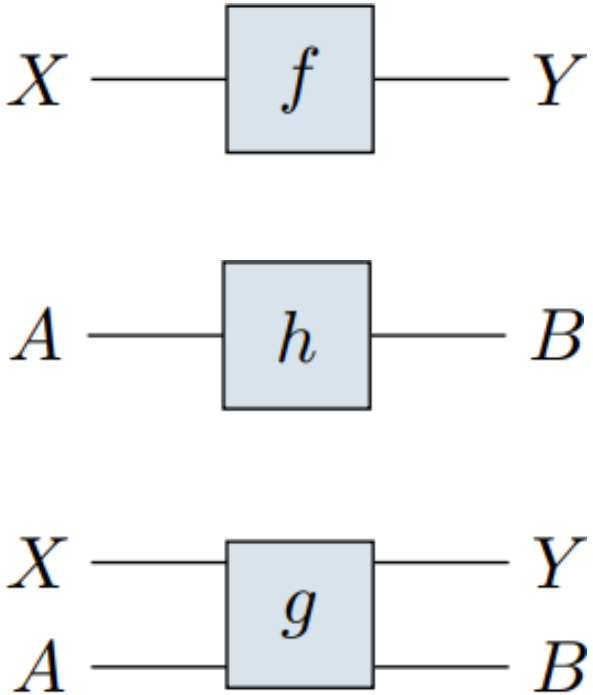

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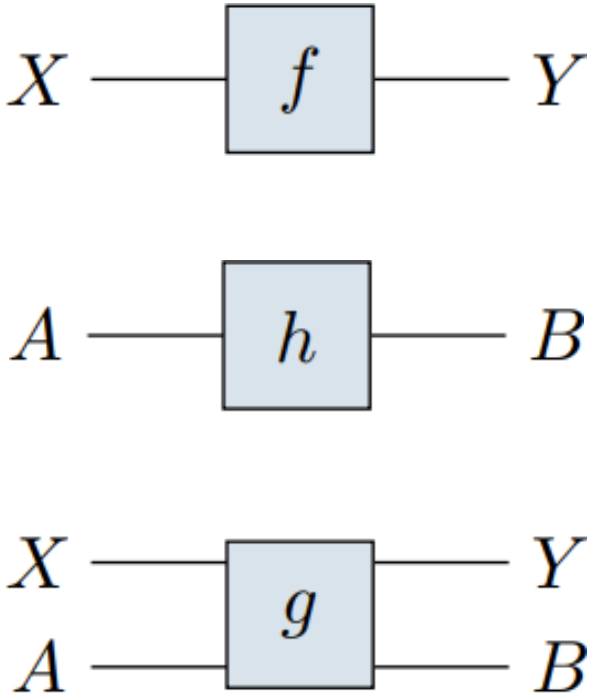

$$g : X \otimes A \to Y \otimes B$$

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Definition

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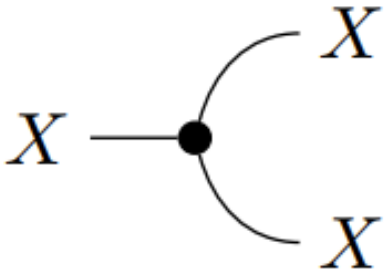

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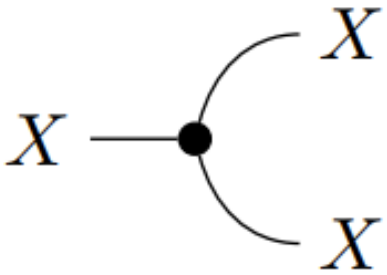

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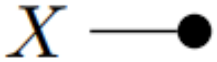

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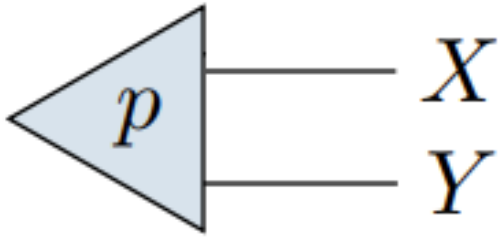

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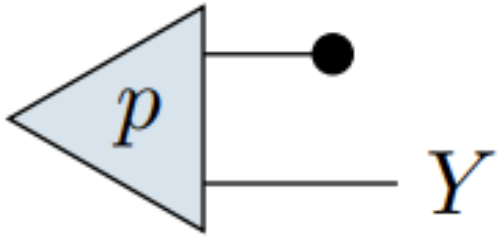
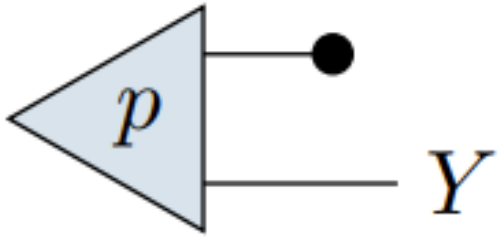


Del

$$\mathbb{P}_Y(y) = \sum_{x \in X} \mathbb{P}_{X,Y}(x, y)$$



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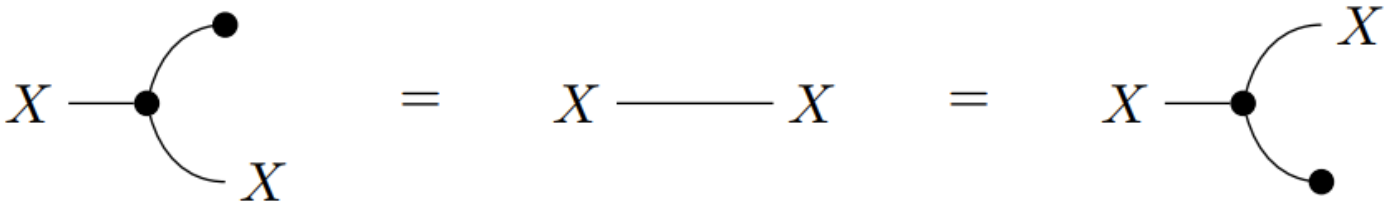

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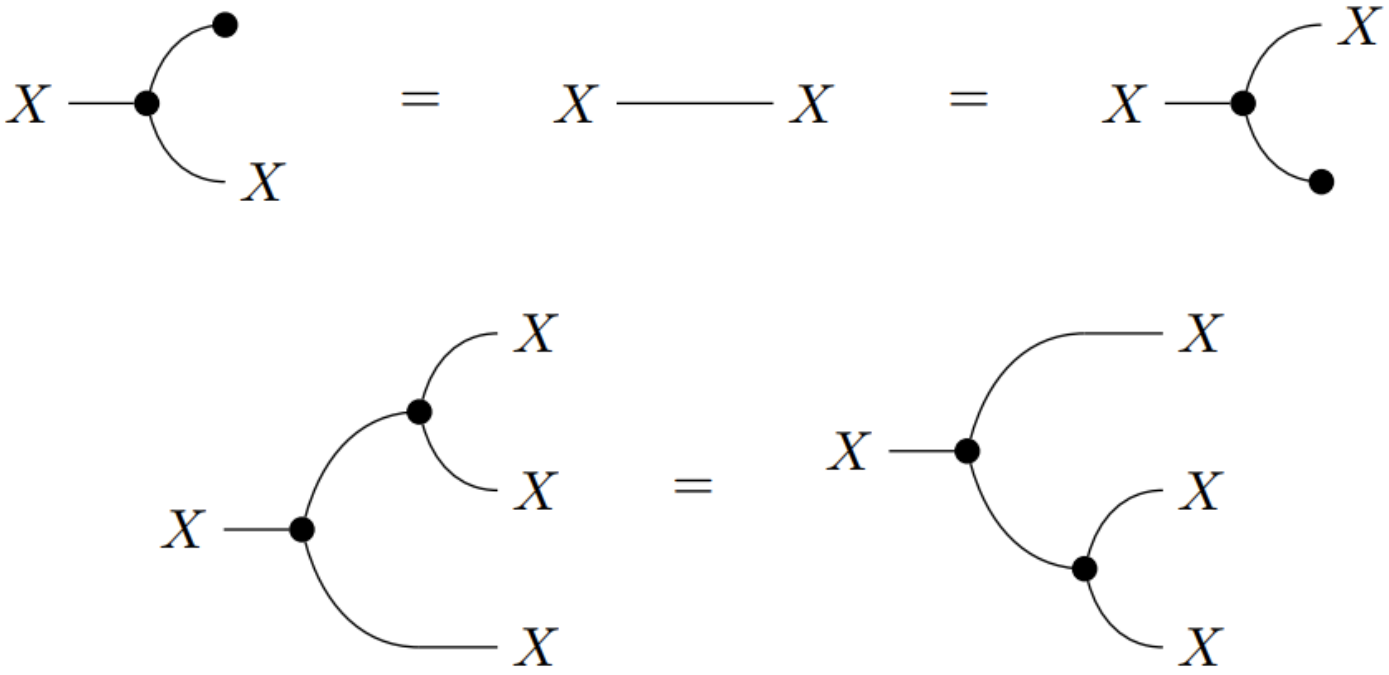

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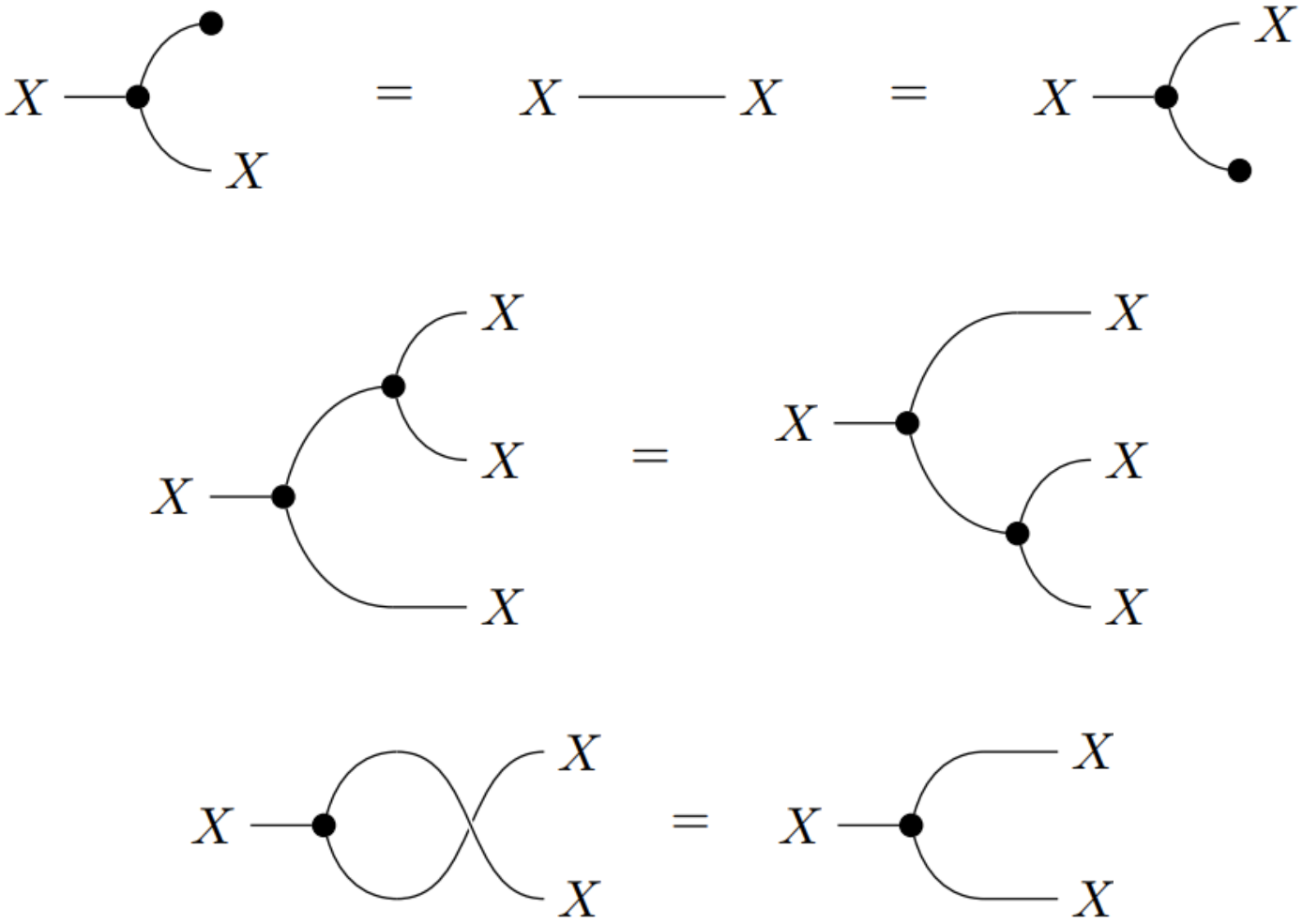
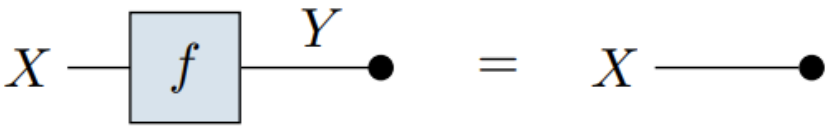


Naturality axiom



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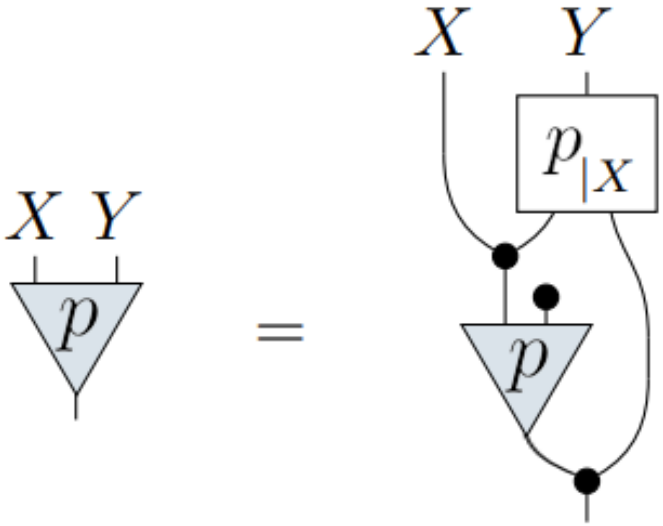

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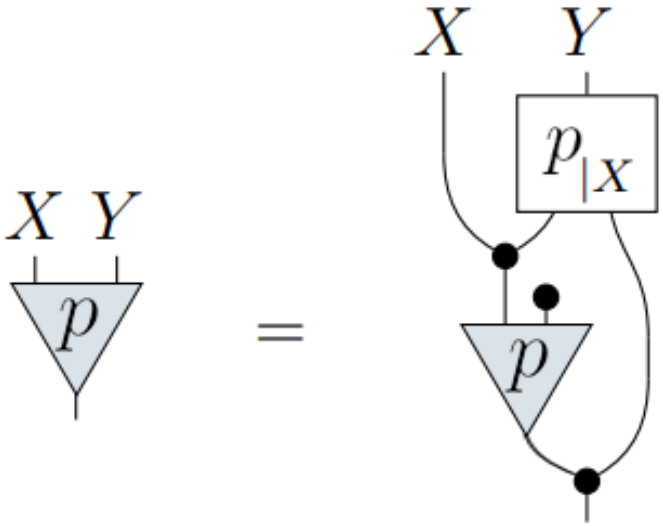

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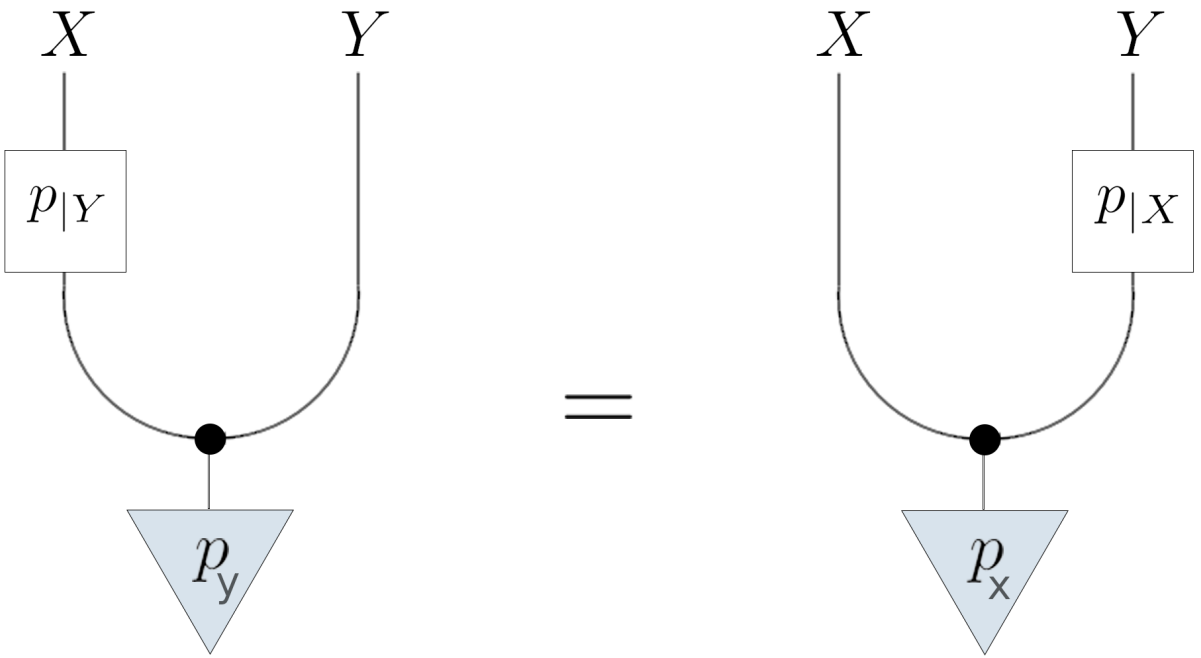

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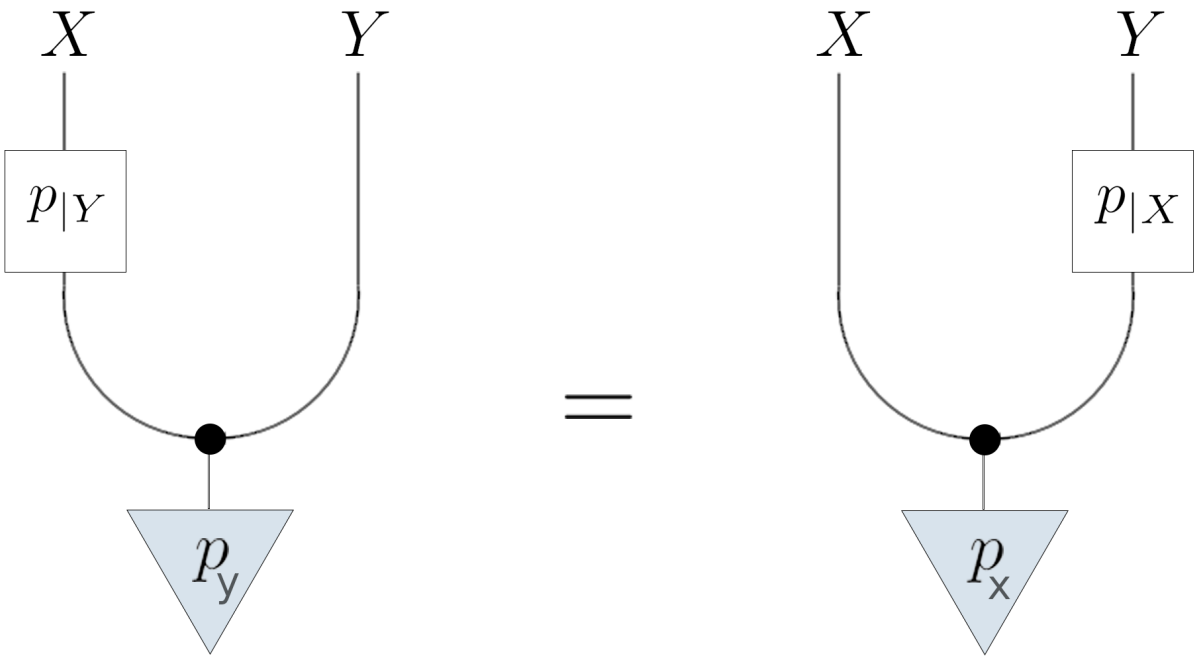

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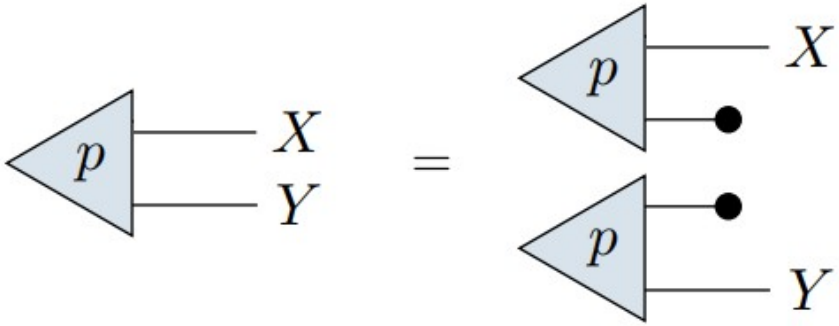

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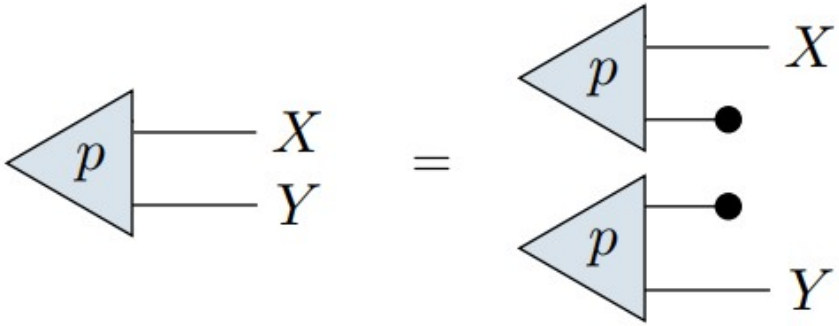

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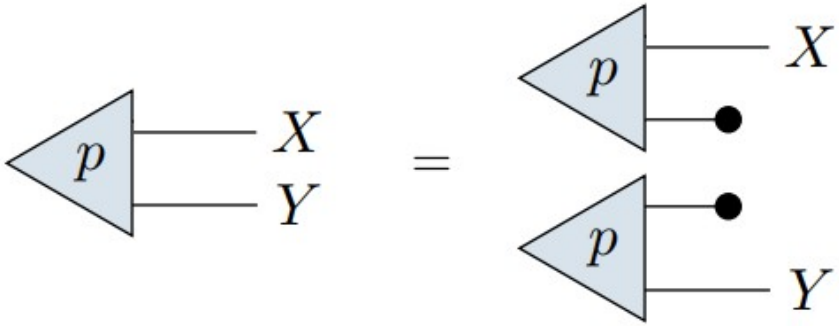

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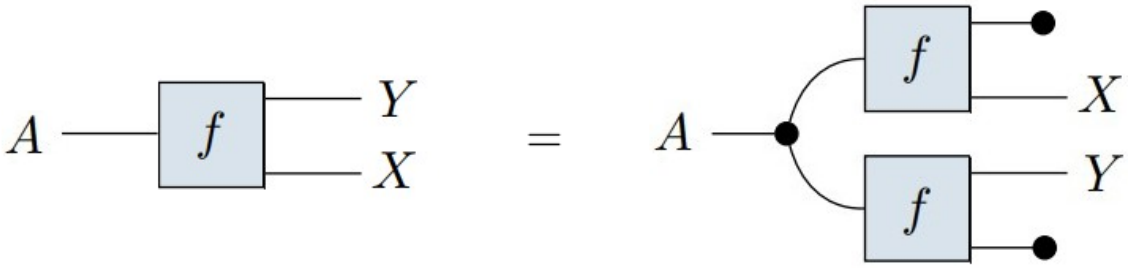

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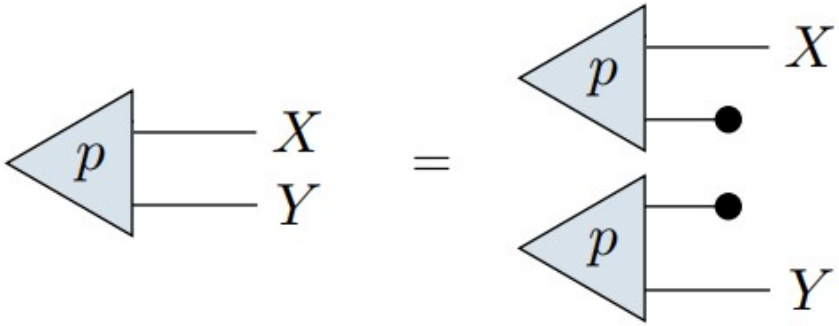

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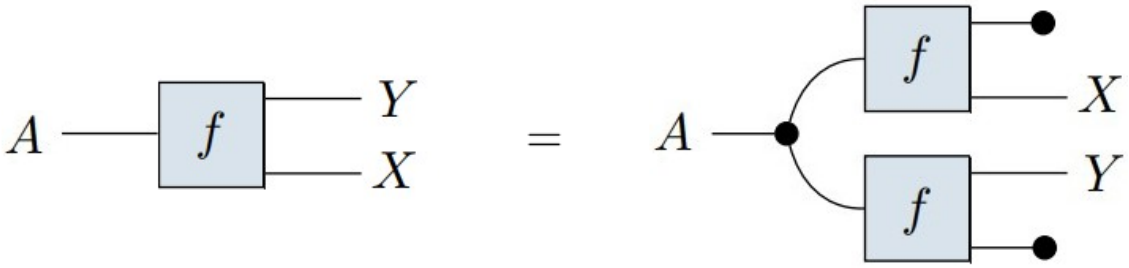

$$\mathbb{P}_{X,Y|\,A}(x, y|a) = \mathbb{P}_X(x|a)\mathbb{P}_Y(y|a)$$

Applications : Sufficient statistics [2]

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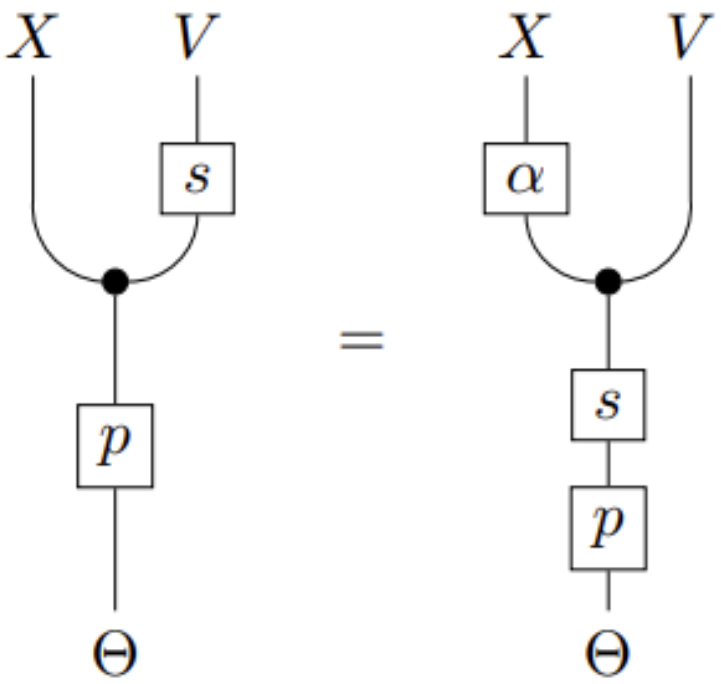
Which means that we have , $\alpha : V \to X$, s.t



Where to go

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- and much more! [7]

References

[1] Brendan Fong. *Causal Theories: A Categorical Perspective on Bayesian Networks*. 2013. eprint: arXiv:1301.6201.

[2] Tobias Fritz. "A synthetic approach to Markov kernels, conditional independence and theorems on sufficient statistics". In: (2019). DOI: 10.1016/j.aim.2020.107239. eprint: arXiv:1908.07021.

[3] Tobias Fritz, Tomáš Gonda, and Paolo Perrone. "De Finetti's Theorem in Categorical Probability". In: (2021). DOI: 10.31390/josa.2.4.06. eprint: arXiv:2105.02639.

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[5] Paolo Perrone. *Markov Categories and Entropy*. 2022. eprint: arXiv:2212.11719.

[6] Dan Shiebler. *Generalized Optimization: A First Step Towards Category Theoretic Learning Theory*. 2021. eprint: arXiv:2109.10262.

[7] Dan Shiebler, Bruno Gavranović, and Paul Wilson. *Category Theory in Machine Learning*. 2021. eprint: arXiv:2106.07032.

Thank You

Questions